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# Agent-Based Model (NetLogo) Report
# Title: Underwater Trash-Collecting Robots
# Group Name: Parallel


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## 1. Introduction

Our NetLogo model focuses on the underwater part of the mission which are the observation ROV (ObsROV) sweeps the seabed and maps litter, while the collection ROV (ColROV) picks up the mapped litter and returns it to the basket. Marine life is included because the robots must work around it without causing harm.

## 2. How the Model Works

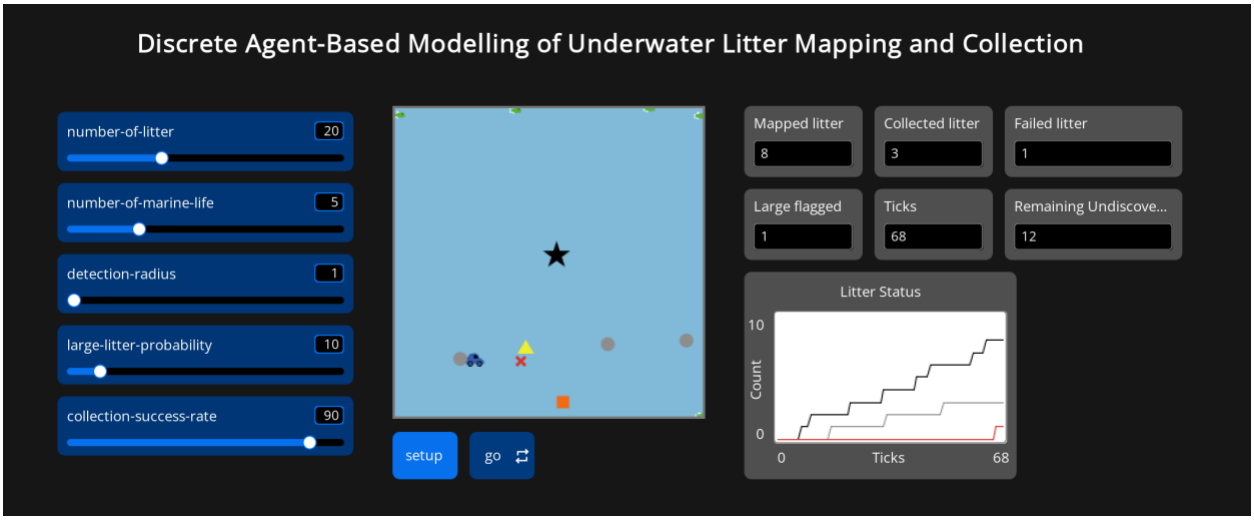



*Figure 2.1 NetLogo Interface of the SeaClear Litter Model*

The seabed is a 20 × 20 patch grid with no wrap-around, and time moves in discrete ticks. In every tick, the observation ROV (ObsROV) acts first, then the collection ROV (ColROV), then the marine life.

| Agent | Represents | Rule in each tick |
|---|---|---|
| ObsROV | miniTortuga | Moves one patch along a lawnmower path starting at (0, 1) and maps hidden litter within detection-radius. Holds position if a fish is within 1 patch. |
| ColROV | Tortuga | Moves one patch towards the nearest mapped small litter. Within 1 patch, picks it up with probability collection-success-rate and returns it to the basket. |
| Litter | Seabed debris | Static. Goes from undiscovered to on-map, then to collected or failed. Large litter is mapped but skipped by the ColROV. |
| Marine life | Fish | Turns randomly by up to 30° and moves one patch. |
| Basket | Collection basket | Fixed at (10, 10). Start point of the ColROV and drop-off point for litter. |

## 3. Results

Figure 2.1 follows one default run. Collection stays close behind mapping, the ColROV delivered its first litter at tick 21 and kept just below the mapped line, so both robots work in parallel. The run ended at tick 372 with 20 litter mapped, 16 collected, 2 failed and 2 large litter flagged. We then ran each setting 100 times with different random seeds and a 1,000-tick limit.

| Setting | Mapped | Collected | Failed | Runs finished | Mission time (ticks) |
|---|---|---|---|---|---|
| Default | 19.4 / 20 | 15.6 | 1.7 | 54% | 419 |
| Detection radius 2 | 20.0 / 20 | 16.0 | 1.9 | 100% | 379 |
| Detection radius 5 | 20.0 / 20 | 15.9 | 2.0 | 100% | 318 |
| 20 marine life | 19.3 / 20 | 15.6 | 1.7 | 53% | 565 |
| Success rate 50% | 19.4 / 20 | 8.7 | 8.7 | 54% | 411 |
| Success rate 100% | 19.4 / 20 | 17.3 | 0.0 | 54% | 415 |
| 50 litter | 48.6 / 50 | 39.2 | 4.4 | 29% | 606 |

Detection range had the biggest effect. At radius 1, only 54% of runs finished. The ObsROV sweeps from row 1, so litter below that row is outside its sensor range, and one missed piece keeps the run going until the tick limit. At radius 2 or more, every run mapped all litter and finished, and mission time fell at radius 5 because litter was found earlier in the sweep. Marine life lengthened the mission because the ObsROV had to hold position more often, and 20 fish added about 159 to 200 ticks. The success rate changed how much litter was collected but not the mission time, since failed litter is not retried. With 50 litter, the mission took 606 ticks, showing that one ColROV becomes the bottleneck as litter increases.

## 4. References

Ilioudi, A., Sosnowski, S., Banken, E., Bevanda, P., Brüdigam, J., Buşoniu, L., Chardard, Y., Delea, C., De Schutter, B., Đuraš, A., Hertel-ten Eikelder, C., Heshmati-Alamdari, S., Maer, V.-M., Palunko, I., Pozniak, I., Prkačin, V., & Tolić, D. (2026). The SeaClear system: An intelligent multi-robot solution for autonomous cleanup of marine debris on the seabed. *Engineering Applications of Artificial Intelligence, 170*, Article 114094. https://doi.org/10.1016/j.engappai.2026.114094

**GitHub Repository:** https://github.com/syedmuhdhafidz/seaclear-netlogo-abm